

Rachel Sunarko

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EDUCATION

Undergraduate in Data Science Technology

Universitas Airlangga | 2023 – 2027

- Faculty Representative – SMARTA 2025 Student Outbound Program to Universiti Malaya
- Presenter – SMARTA 2025 Student Inbound Program (participants from Universiti Malaya & Nanyang Polytechnic)
- Representative – Data Science Technology Major for Faculty Most Outstanding Student Selection (2025)
- Finalist – Most Outstanding Student, Faculty of Advanced Technology and Multidiscipline (2025)

WORK EXPERIENCE

Data Scientist Intern

PT Digital Mediatama Maxima Tbk. | Jul 2026 – Aug 2026

- Executed a data analysis project on DMMX's PPOB (payment point) business to identify key drivers of sales performance, per assigned task brief.
- Generated a synthetic transaction dataset (Python/Pandas) simulating product categories, brands, and agent stores with realistic seasonality and demand patterns.
- Performed exploratory data analysis on revenue trends, category/brand performance, store segmentation, and incentive program effectiveness.
- Built a Power BI dashboard to present analysis findings and an interactive Streamlit dashboard to simulate sales scenarios under configurable business assumptions.

Data Scientist Intern

Center for Environmental, Social, and Governance Studies | Dec 2025 – Present

- Conducted large-scale web scraping and crawling on thousands of PDF documents stored in Google Drive using Python and Google Drive API.
- Processed and transformed unstructured text data using OpenAI API for intelligent extraction, summarization, normalization, and metadata generation.
- Developed scalable document indexing workflows integrated with Zilliz (Milvus Vector Database) to enable semantic search and retrieval across large document repositories.
- Developed an SDG logo detection system by creating and annotating training datasets, training computer vision models in Roboflow, and integrating the detection pipeline into document processing workflows for automated identification of Sustainable Development Goals indicators within PDF documents.

Data Analyst Intern

SDG Center Universitas Airlangga (in collaboration with Surabaya City Government) | Jun 2025 – Sep 2025

- Collected and validated SDGs data for official collaboration reports
- Managed structured datasets for inter-institutional documentation

EXTRA CURRICULAR & CERTIFICATIONS

Deloitte Australia Data Analytics Job Simulation – Forage | Feb 2026

- Completed a Deloitte job simulation involving data analysis and forensic technology.
- Created a data dashboard using Tableau.
- Used Excel to classify data and draw business conclusions.

Tata Data Visualisation: Empowering Business with Effective Insights Job Simulation – Forage | Feb 2026

- Completed a simulation involving creating data visualizations for Tata Consultancy Services.
- Prepared questions for a meeting with client senior leadership.
- Created visuals for data analysis to help executives with effective decision making.

PROJECTS

Orchestra Member ETL Pipeline & Analytics Dashboard

Data Engineering Portfolio Project – Apache Airflow · PostgreSQL · Python · Docker

- Designed and implemented a containerized end-to-end ETL pipeline using Apache Airflow, PostgreSQL, and Python to process orchestra membership data across faculties and instrument sections.
- Architected a Star Schema data warehouse with staging and warehouse layers, separating raw ingestion from cleaned analytical tables to support scalable reporting.
- Developed modular Python ETL modules for extraction, transformation (multi-instrument parsing, phone normalization, deduplication), and idempotent loading via upsert logic.
- Built an Airflow DAG with six sequential tasks, XCom-based metadata passing, retry policies, and pipeline run logging for observability and audit trails.
- Implemented six automated data quality checks covering null validation, duplicate detection, enrollment year integrity, and phone coverage with pipeline-blocking enforcement on critical failures.
- Integrated Metabase as the analytics layer, delivering an interactive dashboard to visualize member composition by section, instrument distribution, faculty breakdown, and year-over-year recruitment trends.
- Wrote 20+ unit tests using pytest to validate transformation logic independently of the database environment.

E-Clinic Lab – Healthcare Laboratory Information System

Final Project – Advanced Database · MySQL · Pentaho Data Integration · OLAP

- Designed and implemented a relational database system using MySQL to support transactional operations across healthcare laboratory branches.
- Developed normalized database schemas and data models to ensure data consistency, integrity, and efficient query performance.
- Architected an OLTP-OLAP environment by separating operational databases from analytical data warehouses to support reporting and decision-making.
- Built end-to-end ETL pipelines using Pentaho Data Integration (PDI) to extract, transform, validate, and load data from transactional systems into the data warehouse.
- Implemented data quality checks and transformation rules during ETL processes to improve data accuracy and reliability.
- Developed analytical dashboards powered by the OLAP warehouse to monitor laboratory performance, operational trends, and branch-level metrics.
- Optimized SQL queries and warehouse structures to improve reporting performance and support scalable analytical workloads.

Document Checker – Multi-Model Text Similarity Implementation on NLP-based Document Checker Web-App

Final Project – Natural Language Processing · Python · NLP · Transformer Embeddings

- Developed a web-based document comparison system capable of analyzing 2-10 documents simultaneously across multiple formats (PDF, DOCX, TXT, and images via OCR).
- Implemented an integrated NLP pipeline including text extraction, preprocessing, and pairwise similarity computation.
- Applied statistical, textual, and transformer-based embeddings methods to compute similarity across documents.

PackMe – Object Size Prediction and Packaging Recommendation Web-App

Final Project – Algorithm and Programming II · HTML · CSS · JavaScript

- Designed and developed a responsive landing page for Pack Me, a web app that predicts package sizes based on user input.
- Implemented using HTML, CSS, and JavaScript with clean, modern UI principles.
- Focused on layout design and smooth user interaction.
- Collaborated with backend teammates to align front-end components with prediction logic.

PROFESSIONAL & ORGANIZATIONAL EXPERIENCE

Teaching Assistant – Algorithmic Programming I

Fakultas Teknologi Maju dan Multidisiplin, Universitas Airlangga | 2025

- Assisted 60+ students in Python programming and debugging
- Guided students in core concepts: variables, control structures, loops, functions, and data structures
- Provided one-on-one mentoring during practical sessions

Researcher – IRIS | 2023 – 2024

- Collaborated on financial sentiment analysis using Yahoo Finance news data
- Performed web scraping using Selenium to collect dynamic headlines and timestamps

Head of Student Resource Development

Airlangga Orchestra | 2025

Staff – Ministry of Science, Research & Technology

BEM FTMM, Universitas Airlangga | 2024

- PIC – Arsip Karya Mahasiswa (ARKAMA)
- Vice KOC – Ruang Apresiasi Mahasiswa (RUMA)
- Event Coordinator – Workshop KIM & PKM 2024

Creative Lead – BINARY 4.0 | 2024

- Led design, publication, decoration, and documentation for event branding
- Developed Instagram feeds, logo modifications, and event visuals
- Produced animated twibbon, camp logo, mentor reveal video, and teaser

Secretary – Airlangga Orchestra Internship (ARSIP) | 2024

Staff – Student Resource Development – Airlangga Orchestra | 2024

COMPETITIONS

[Faculty, 2025] Finalist – Most Outstanding Student of Faculty of Advanced Technology and Multidiscipline

[National, 2025] 3rd Place – National Data Analytic Competition of EPSILON 2025 by Universitas Negeri Padang

- Created a data-driven analysis project titled "Wawasan Pendidikan untuk Masa Depan yang Lebih Cerdas: Analisis Kesenjangan dan Strategi Pendidikan Antar Provinsi di Indonesia".
- Utilized Exploratory Data Analysis (EDA), Spearman correlation, K-Means clustering, and OLS regression to analyze educational and socio-economic data across 34 Indonesian provinces.
- Segmented provinces into four distinct archetypes based on their unique challenges, identified key factors to uncover their significant influence on educational outcomes, and provide evidence-based insights to develop a cluster-based policy framework.

[National, 2025] 3rd Place – Data Science Category of IMILKOM USU IT Competition 2025 by Universitas Sumatera Utara

- Created a clustering-based analysis project titled "Analisis Penerapan Strategy on Education for Health and Well-being UNESCO dengan Indikator SDG di Indonesia (2012–2023)".
- Employed PCA and K-Means clustering on education and health data to uncover region-based implementation patterns of UNESCO strategies.
- Provided actionable insights to support policy-making efforts for enhancing student well-being and inclusive learning environments.

[National, 2025] Finalist (Top 5) – Data Analytics Category of TECHFEST 2025 by Universitas Bina Nusantara

- Developed an exploratory data analysis project titled "Studi Eksploratif Efektivitas Penyaluran TKDD: Analisis Perbandingan Pagu, Realisasi, Dan Implikasinya Terhadap IPM Provinsi Di Indonesia Tahun 2023".
- Analyzed, interpreted, and discovered powerful insights from fiscal and socioeconomic data by performing univariate, bivariate, and multivariate statistical analysis to assess their relation to Human Development Index (IPM).
- Generated policy-driven insights to guide targeted budget interventions and improve regional development outcomes across Indonesian provinces.

[National, 2024] Finalist (Top 10) - Infographic Category of 4C National Competition by Universitas Brawijaya. Designed an infographic envisioning the future of railway services through digital transformation for enhanced public service.

[National, 2024] Best Infographic Visual Quest of Airnology 3.0 by Universitas Airlangga. Created a data-driven visual narrative on green hydrogen as a clean energy solution for Indonesia's air quality.

SKILLS

- **Hard Skills:** Programming Languages (Python, C++, SQL, R, HTML, CSS, JavaScript) • Data Handling & Analysis (Google Colaboratory, Jupyter Notebook, R Studio, Visual Studio Code, Tableau, Microsoft Excel) • Data Engineering & Databases (SQL, Database Design & Data Modelling, ETL Pipelines, Pentaho Data Integration, Vector Databases) • Machine Learning & NLP • Programming & Web Development (Full-Stack Web Development, Backend API Integration) • GIS Software (QGIS) • AI Tools & Frameworks (Data Processing Automation, OpenAI API Integration) • Version Control & Platforms (GitHub, Git) • Design & Multimedia Tools (Adobe Illustrator, Adobe Premiere Pro, Adobe After Effects, Adobe InDesign, Canva, Figma) • Office Tools (Microsoft Word, Microsoft PowerPoint, Microsoft Excel)
- **Language Skills:** Indonesia (Native), English (ELPT: 597 | TOEFL ITP Prediction: 630)
- **Music & Performance:** Pianist, Cellist – Airlangga Orchestra (Performed at MAWAPRES Final 2024, 13th ICAS Dinner, SOD Fest Vol. 6, PIMNAS 2024, and other events)
- **Soft Skills:** Teamwork & Leadership, Logical & Innovative Thinking, Leadership, Strong Work Ethic, Adaptability, Problem-Solving, Work Well Under Pressure